

(a) Co-training Pipeline

 Vision-Language- Action Data



 Real-world 3D Reasoning Data



VLM Backbone



Action Head

a_1 a_2 $\dots$ a_T

Explicit 3D Reasoning Text Generation

The 3D **location** of the bus is $(-4.3, 1.9, 17.7)$...

(b) Action Prediction by co-training with 3D data only

Action Prompt

Action Prediction

GAP

3D Reasoning Prompt

3D Spatial Reasoning

(c) Ours: Action Prediction via 3D Spatial Reasoning

Action Prompt

Action Prediction

Injection

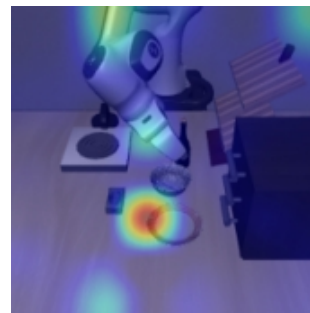
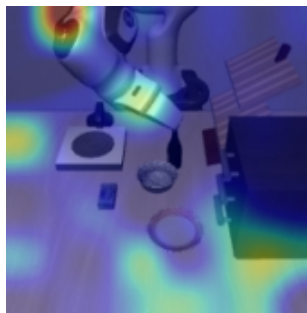
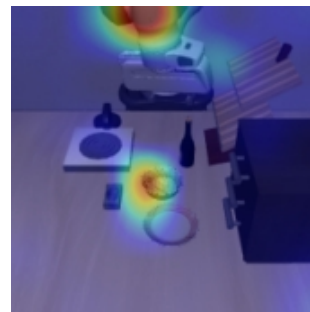
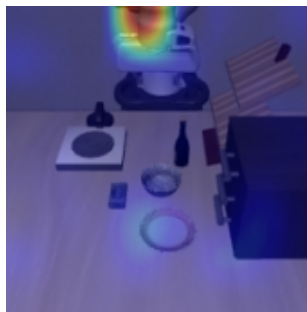
3D Reasoning Prompt

3D Spatial Reasoning

(d) LIBERO Benchmark

Co-train Only

Ours



(e) LIBERO-PLUS Benchmark

Co-train Only

Ours

